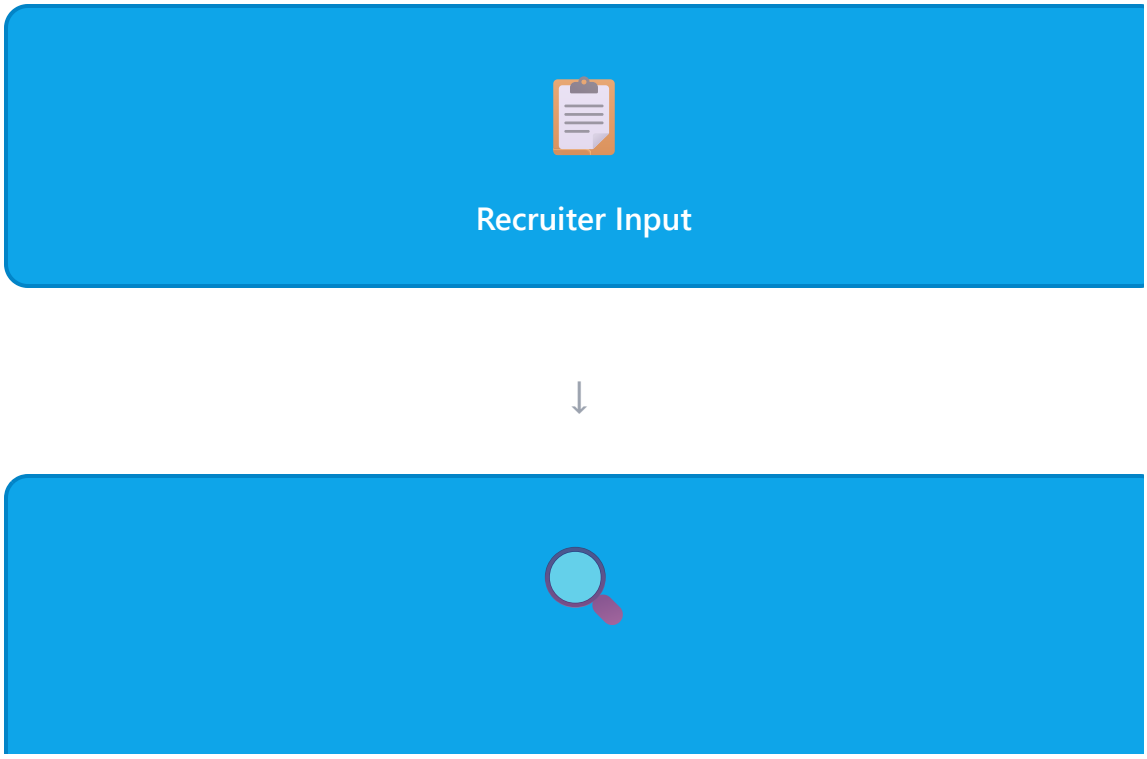
Gainday AI Engine

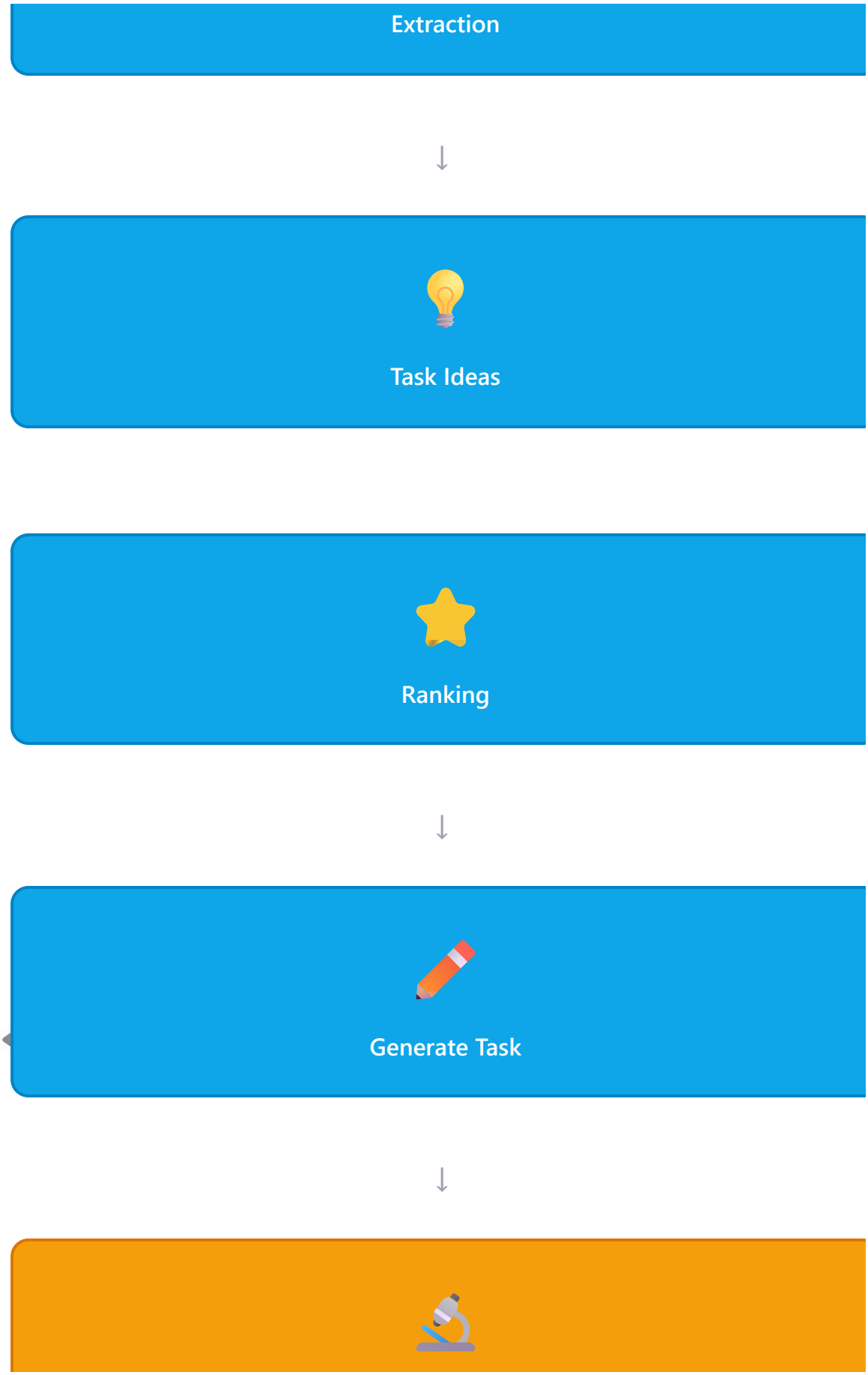
How the AI Engine Generates and Grades Job Simulations

Part 1: Generation & Regeneration

How the AI engine creates realistic job simulation tasks from a recruiter's job posting

The Generation Pipeline




Extraction
Task Ideas
Ranking
Generate Task



✓ Yes: Persist

✗ No: Regenerate

📌 The Steps Explained

📋 Step 1: Recruiter Input

A recruiter provides three things:

- **Job description** - What the role does
- **Required skills** - What we should test
- **Business problem** (optional) - A real problem they want solved

🔍 Step 2: Extraction

The AI analyzes the input and extracts:

- **Category** - Is this Sales? Finance? (routes to the right module)

- **Intent** - The specific skills to test
- **Problem** - The actual business problem (if provided)

💡 Step 3: Generate Task Ideas (15 Candidates)

Instead of creating one task, the AI generates **~15 possible task ideas**. This "overgenerate and rank" approach ensures variety and quality. Why? Because picking the best idea from many options is more reliable than trying to create the perfect task immediately.

⭐ Step 4: Rank & Select Top 4

The AI scores each of the 15 ideas (0-10 points) on three criteria:

- Does it match the required skills? (Intent alignment)
- Is it appropriate for this role? (Category alignment)
- Does it incorporate the business problem? (Problem alignment, if provided)

Result: The top 4 ideas move forward.

🖍️ Step 5: Full Task Generation

For each of the 4 selected ideas, the AI creates:

- **Task title & scenario** - The context and what the candidate must do
- **Response format** - Whether it's a written answer, a calculation, a ranking, etc.
- **Metadata** - Labels for how the frontend should display this task

🔬 Step 6: Quality Critique

Before a task is shown to candidates, it's automatically checked:

- ✓ **Is it novel?** (Not too similar to existing tasks)

- ✓ **Is it relevant?** (Actually tests what was requested)
- ✓ **Is it sound?** (Follows domain best practices)

🔄 Regeneration Loop (When Quality Fails)

♻️ If a task fails critique...

The system doesn't start over.

Instead, it picks the next-best idea from the original 15 candidates and generates that instead. This happens **up to 3 times.**

If all 3 attempts fail: The task is flagged for human review. It doesn't get shown to candidates until an admin approves it.

💡 **Why this design?** Generating one task at a time is risky (you might get duplicates or off-topic tasks). Generating 15 and ranking them is much more reliable. If one ranking fails, there are backups waiting.

📊 Part 2: Grading

How the AI engine scores candidates' answers fairly and consistently

The Grading Pipeline



Candidate Submits

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Anchors Ready

↓

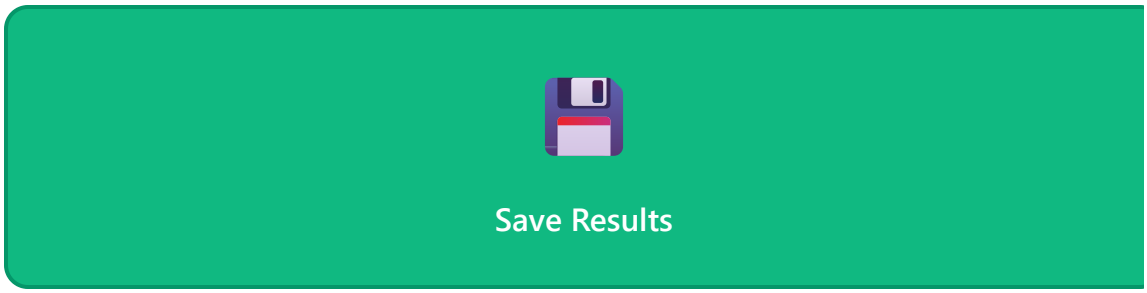


Grade Each Task

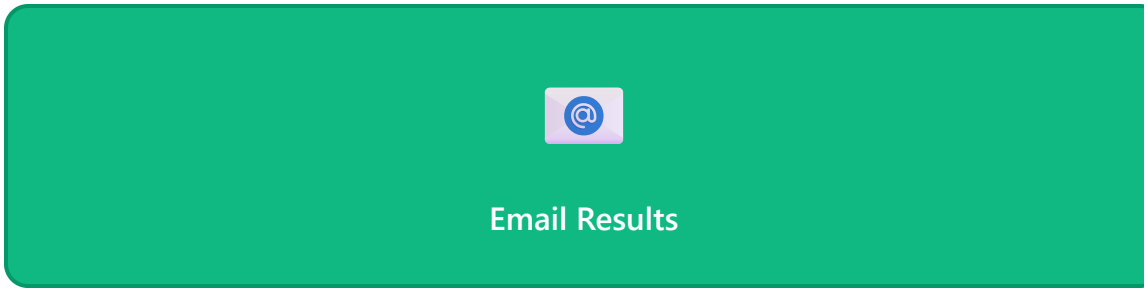


Roll Up Scores

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📌 The Steps Explained

✍️ Step 1: Candidate Submits Answers

A candidate completes all tasks in a simulation and submits their answers. The system receives them and immediately starts grading in the background—the candidate doesn't have to wait.

🎯 Step 2: Prepare Anchor Responses (Reference Standards)

Before grading a task, the system needs **reference answers** to calibrate scoring against. Think of these as:

- **A perfect answer (100/100)**
- **A strong answer (80/100)**
- **An okay answer (60/100)**
- **A weak answer (40/100)**
- **A poor answer (20/100)**

Important: These anchors are created **only the first time** a candidate answers a task. After that, they're reused for all future candidates on the same task. This saves cost and ensures consistency.

📝 Step 3: Grade Each Task

For **each task**, the AI compares the candidate's answer against the 5 anchor responses and scores them on 4 dimensions:

🎯 Problem-Solving

Did they identify and solve the core problem? Does their approach make sense?

💼 Judgment & Execution

Did they make smart decisions? Were they thorough and methodical?

✍️ Written Communication

Is their answer clear, well-structured, and easy to follow?

🏆 Commercial/Domain Awareness

Do they understand the business or industry context? Is their thinking practical?

💡 **Why 5 anchors, not just 2?** Comparing to only "good" (100) and "bad" (0) doesn't help grade the middle—where most real answers fall. With anchors at 20, 40, 60, 80, and 100, the AI can accurately distinguish between a 55 and a 65 (the hardest and most important distinction in hiring).

📈 Step 4: Roll Up Scores

Each task is now scored (0-100) on all 4 dimensions. The system then:

- **Averages each dimension** across all tasks (e.g., problem-solving scores: 80, 70, 90 → average 80)
- **Weights them** by importance:

  • Problem-Solving: **40%** (most important)
  • Judgment & Execution: **20%**
  • Written Communication: **20%**
  • Commercial/Domain Awareness: **20%**

Final score: 0-100 (where 100 is perfect)

💾 Step 5: Save Results

The system stores:

- The candidate's overall score (0-100)
- Their score on each of the 4 dimensions (0-100)
- Detailed breakdowns per task
- A summary of their strengths and weaknesses

📧 Step 6: Email Results

The candidate receives an email showing:

- Their overall score (out of 100)
- Breakdown by the 4 dimensions
- A summary for each task ("You did well here because...")
- Specific feedback grounded in their actual answers

💡 **Why this design?** By anchoring to reference answers instead of absolute rules, the system is fair even as tasks change or improve. It's also consistent—every candidate on the same task is compared using the same reference points, regardless of when they submit.

🔄 Generation vs. Grading: Side by Side

⚡ Generation

- ✓ Runs **before** any candidate applies
- ✓ Recruiter uploads job description
- ✓ System creates 4 tasks automatically
- ✓ Quality-checked by critique system
- ✓ Shown to **all candidates** identically
- ✓ Goal: Create fair, relevant, quality tasks

🏁 Grading

- ✓ Runs **after** candidate submits
- ✓ Candidate submits their answers
- ✓ System grades all 4 tasks automatically
- ✓ Uses anchor references for consistency

- ✓ Scored on 4 important dimensions
- ✓ Goal: Score fairly and give actionable feedback

💎 Why This Approach?

✨ Quality

- • Generate many ideas, pick the best
- • Automatic quality checks prevent bad tasks
- • Regeneration ensures variety
- • Admin review for edge cases

⚖️ Fairness

- • All candidates see the same tasks
- • Anchor references ensure consistent scoring
- • First-time anchor generation ensures fairness
- • Weighted dimensions reflect real hiring needs

⏱️ Efficiency

- • No manual task writing by recruiters
- • Anchors generated only once (cost savings)
- • Fully automated background grading
- • Scales to hundreds of candidates

📊 Insights

- • Candidates get detailed feedback per task
- • Recruiters see standardized scoring
- • Capability scores track growth over time
- • Data-driven hiring decisions

Gainday AI Engine Documentation

For more details, see: gainday-generation-pipeline.md & gainday-grading-pipeline.md